Can 3D JEPA Segment? Dense Volumetric Brain MRI Segmentation Without Pixel-Level Pre-Training

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Abstract

Joint-Embedding Predictive Architectures (JEPA) learn abstract representations by predicting masked regions in latent feature space rather than reconstructing raw pixel or voxel intensities. This design raises a fundamental question for medical imaging: *can an architecture trained without voxel-level reconstruction perform dense, per-voxel multi-compartment volumetric segmentation?* We address this through a controlled benchmark on the BraTS 2024 Adult Glioma dataset evaluating Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET) segmentation under the clinical composite three-region protocol. We assess a teacher-free single-encoder **3D VisReg JEPA** with multi-scale FPN and UNETR hybrid decoders against matched from-scratch Vision Transformer controls and established 3D convolutional baselines (Residual UNet and deeply supervised 3D nnU-Net). On held-out test volumes ($N = 242$), 3D VisReg JEPA demonstrates that latent prediction successfully supports dense segmentation, reaching 69.66% mean Dice (84.11% WT, 62.56% TC, 62.30% ET) with a hybrid UNETR decoder, outperforming matched from-scratch ViT-UNETR (68.19%, $+1.47$ pp). Furthermore, completely freezing the pre-trained ViT backbone achieves 69.34% mean Dice (83.29% WT, 62.38% TC, 62.35% ET) with a hybrid UNETR probe and 63.18% with a pure latent multi-scale FPN probe, demonstrating that self-supervised latent prediction learns rich, spatially decodable tumor representations prior to downstream adaptation. However, controlled comparisons establish three critical boundary conditions: (1) under full supervision (100% labels), 3D nnU-Net maintains the performance ceiling (70.86%) due to localized convolutional inductive biases, closely followed by VisReg UNETR (69.66%); (2) under extreme and moderate annotation scarcity (1% to 10% labels), VisReg JEPA establishes top overall performance across the entire benchmark (34.32% mean Dice at 1% for frozen probe, beating nnU-Net at 20.69% by $+13.63$ pp; and 54.45% at 5%, beating nnU-Net at 47.06% by $+7.39$ pp and scratch ViT at 49.79% by $+4.66$ pp); and (3) under missing pulse sequence triage, VisReg JEPA demonstrates superior resilience on FLAIR-only inputs (44.61% mean Dice vs. 39.79% for nnU-Net), while pre-training shields UNETR against severe single-sequence degradation on T1c-only (49.22% vs. 36.55% for scratch UNETR, a $+12.67$ pp robustness margin). Finally, spectral analysis confirms that VisReg JEPA expands effective latent dimensionality ($S^2 = 69.13$, 18.0% capacity) compared to severe dimensional collapse in from-scratch transformers ($S^2 = 30.95$, 8.1%). These findings demonstrate that 3D JEPA provides compelling sample efficiency and feature regularized stability without requiring voxel reconstruction.

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1 Introduction

Automated volumetric segmentation of pathological brain lesions in multi-parametric Magnetic Resonance Imaging (MRI)—such as gliomas imaged via co-registered T1, post-contrast T1 (T1c), T2, and Fluid-Attenuated Inversion Recovery (FLAIR) acquisitions—is a cornerstone of neuro-oncology diagnosis, surgical boundary planning, and conformal radiotherapy targeting [Baid et al., 2024, Menze et al., 2014]. Fully supervised 3D convolutional architectures, notably 3D U-Net [Ronneberger et al., 2015, Çiçek et al., 2016], V-Net [Milletari et al., 2016], and 3D nnU-Net [Isensee et al., 2021], have long established the performance standard in volumetric medical image segmentation. Their success is commonly attributed to high-resolution isotropic spatial feature hierarchies and dense horizontal skip connections, which route low-level edge, gradient, and texture cues directly from early encoder layers into the volumetric decoder, ensuring precise spatial localization of anatomical boundaries.

However, training supervised 3D deep models from random initialization requires hundreds of densely contoured 3D volumetric masks delineated across dozens of contiguous axial slices by expert neuroradiologists. Manual voxel contouring is prohibitively expensive, labor-intensive, and subject to marked inter-observer discordance [Menze et al., 2014]. Self-Supervised Learning (SSL) offers an appealing paradigm to alleviate annotation bottlenecks by pre-training representation backbones on unannotated imaging cohorts prior to downstream supervised adaptation.

Generative self-supervised approaches, such as 3D Masked Autoencoders (3D MAE) [He et al., 2022, Feichtenhofer et al., 2022], learn representations by masking spatial voxels or patches and training an expansive decoder to reconstruct raw voxel intensities in image space. In multi-modal brain MRI, however, high-frequency voxel variance is heavily contaminated by non-biological acquisition artifacts: radiofrequency thermal noise (characterized by non-central Rician distributions [Gudbjartsson and Patz, 1995]), scanner-dependent B_1 magnetic field inhomogeneities, and microstructural tissue speckle. Devoting model capacity to 3D voxel-level reconstruction forces the network to allocate expressive resources to uninformative acquisition noise rather than underlying anatomical morphology and pathological lesion boundaries.

The 3D JEPA Paradigm and the Voxel Paradox To overcome the limitations of pixel and voxel reconstruction, Joint-Embedding Predictive Architectures (JEPA) [Assran et al., 2023, LeCun, 2022] operate strictly within latent feature space. Instead of reconstructing masked voxels, an online 3D context encoder processes visible volumetric tokens, and a lightweight predictor network is trained to forecast the latent representations of masked 3D target blocks. By omitting a pixel-level decoder entirely, JEPA is designed to learn abstract semantic representations and spatial contextual relationships while discarding high-frequency noise.

This foundational design choice leads directly to a central research question and an apparent scientific paradox:

Can 3D architectures designed explicitly to avoid voxel-level reconstruction perform dense, per-voxel volumetric segmentation across multiple anatomical tumor compartments?

At first glance, the objectives appear fundamentally incompatible. Dense volumetric segmentation requires voxel-accurate delineation of infiltrative tumor boundaries across 3D space, whereas JEPA's objective explicitly incentivizes the encoder to discard high-frequency spatial variation in favor of abstract semantic representations. If low-level spatial details are discarded during pre-training, can the learned representations support downstream boundary delineation across distinct tumor sub-compartments?

Scope and Contributions In this work, we investigate this question through a controlled empirical study comparing 3D Joint-Embedding Predictive Architectures against standard supervised 3D convolutional baselines in multi-modal volumetric brain MRI segmentation under the clinical BraTS 2024 GLI three-region protocol: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET). Rather than relying on early dual-encoder JEPA variants that require heuristic momentum teachers, we evaluate **3D VisReg JEPA** [Wu et al., 2026], a single-encoder formulation that prevents latent representation collapse through decoupled optimal transport feature regularization. To systematically probe the architectural bottleneck of Vision Transformers in dense volumetric prediction, we evaluate two decoder paradigms: a hierarchical multi-scale 3D Feature Pyramid Network (FPN) and a convolutional skip-connection hybrid (UNETR Hybrid) [Hatamizadeh et al., 2022], paired with matched from-scratch ViT controls and frozen encoder representation probes.

Our contributions are fivefold:

1. **First Multi-Region Benchmark for 3D Latent JEPA**: We evaluate single-encoder 3D VisReg JEPA across the community-standard BraTS 2024 GLI three-region protocol (Whole Tumor, Tumor Core, Enhancing Tumor) on held-out test volumes ($N = 242$). We confirm the clinical difficulty hierarchy (Dice(WT) $>$ Dice(TC) $>$ Dice(ET)) across all architectures and quantify the performance gap relative to deeply supervised 3D CNNs (84.47% vs. 84.11% WT; 64.35% vs. 62.56% TC; 63.76% vs. 62.30% ET).
2. **High-Performing Frozen Encoder Representation Probing**: We evaluate frozen backbone representation quality prior to any fine-tuning. Completely freezing the 3D ViT backbone achieves 69.34% mean Dice (83.29% WT, 62.38% TC, 62.35% ET) with a hybrid UNETR probe, outperforming from-scratch training (68.19%, $+1.15$pp) and closely approaching end-to-end fine-tuning (69.66%), proving that latent masked pre-training natively discovers localized spatial representations that transfer to multi-compartment segmentation without gradient backpropagation through the encoder.
3. **Significant Label-Efficiency Gains in Data Scarcity**: Under extreme data scarcity (1% labels, 11 patient scans), the frozen VisReg UNETR probe achieves the top performance across the entire benchmark (34.32% mean Dice), outperforming deeply supervised 3D nnU-Net (20.69%) by $+13.63$pp and standard 3D Residual UNet (4.29%) by $+30.03$pp. At 5% data (58 scans), VisReg UNETR reaches 54.45% mean Dice, beating nnU-Net (47.06%, $+7.39$pp) and scratch ViT (49.79%, $+4.66$pp), and maintaining this lead at 10% labels (57.68% vs. 50.66% for nnU-Net).
4. **Controlled Pre-Training vs. From-Scratch Equivalence on Full Data**: By evaluating strictly matched from-scratch ViT controls, we show that under abundant annotations (100% labels), VisReg JEPA pre-training provides consistent gains over random weight initialization (69.66% vs. 68.19% for UNETR; 67.05% vs. 66.13% for FPN), while convolutional inductive biases in 3D nnU-Net preserve the overall performance ceiling (70.86%).
5. **OOD and Missing Modality Robustness**: We systematically characterize model robustness across synthetic acquisition perturbations and missing-sequence triage. While fine-tuned architectures tolerate additive Rician noise and B_1 bias fields (< 0.7pp drop, with frozen probes dropping ≤ 2.4pp), VisReg JEPA UNETR establishes top overall performance on emergency FLAIR-only acquisition (44.61% mean Dice vs. 39.79% for nnU-Net), while pre-training shields UNETR against severe single-sequence degradation on T1c-only (49.22% vs. 36.55% for scratch UNETR, a $+12.67$pp advantage).

2 Background: Progression of the JEPA Family

Self-supervised non-contrastive learning in latent space faces a fundamental degenerate failure mode: *latent representation collapse*, where the encoder maps all input patches to a constant vector, driving prediction loss to zero without learning informative representations [Bardes et al., 2022, Zbontar et al., 2021]. The evolution of the JEPA family reflects progressive advances in preventing this collapse.

Dual-Encoder Architectures with EMA Teachers The foundational Image-JEPA (I-JEPA) framework [Assran et al., 2023] prevents representation collapse by maintaining two separate parameter sets: an online context encoder E_θ updated via gradient descent, and a target teacher encoder $E_{\bar{\theta}}$ updated via an Exponential Moving Average (EMA) of online weights:

$$\bar{\theta}_t \leftarrow m\bar{\theta}_{t-1} + (1-m)\theta_t, \qquad m \in [0.996, 1.0] \tag{1}$$

Combined with stop-gradient operators on target representations, the slow evolution of the teacher prevents instantaneous representational collapse [Grill et al., 2020]. However, in 3D volumetric medical imaging, the dual-encoder EMA framework introduces acute drawbacks:

1. **GPU Memory Footprint**: Storing an identical copy of the 3D vision backbone doubles parameter memory during pre-training, severely restricting 3D subvolume batch sizes on memory-constrained GPUs.
2. **Heuristic Scheduling**: Training stability depends heavily on empirical cosine scheduling of teacher momentum $m_t = m_{\text{end}} - (m_{\text{end}} - m_{\text{start}}) \cdot \frac{1}{2}(1 + \cos(\pi t / T_{\text{anneal}}))$ ($m \in [0.996, 1.0]$); improper schedules can induce representation destabilization or asymptotic collapse.
3. **Implicit Mechanism**: Rather than optimizing an explicit objective that penalizes low representation rank or collapsed feature geometry, non-collapse under EMA updating is an emergent optimization phenomenon.

LeJEPA (SIGReg): Teacher-Free Sketched Gaussian Regularization To eliminate the EMA teacher, LeJEPA [Balestriero and LeCun, 2025] introduced Sketched Isotropic Gaussian Regularization (**SIGReg**), enabling single-encoder JEPA training. By the Cramér-Wold theorem [Cramér and Wold, 1936], a multivariate distribution on $\mathbb{R}^D$ is uniquely determined by the family of its 1D marginal projections onto the unit hypersphere $\mathbb{S}^{D-1}$. SIGReg samples random 1D projections and penalizes the discrepancy between the empirical characteristic function (ECF) of projected features and the standard normal characteristic function via the Epps–Pulley test statistic [Epps and Pulley, 1983]. While SIGReg removes the momentum teacher, it couples representation location, coordinate scale, and higher-order distribution shape into a single non-parametric test statistic, which can complicate optimization dynamics when 3D features require simultaneous scaling and de-correlation.

3D VisReg: Decoupled Optimal Transport Regularization Variance-Invariance-Sketching Regularization (**VisReg**) [Wu et al., 2026] refines teacher-free statistical regularization by formally decoupling three geometric degrees of freedom in representation space: (1) batch centering, (2) coordinate scale regularization, and (3) distributional shape regularization via 1D Sliced-Wasserstein Distance (SWD) [Bonneel et al., 2015, Kolouri et al., 2019, Villani, 2009]. Importantly, while foundational single-encoder non-contrastive methods such as LeJEPA [Balestriero and LeCun, 2025] and 2D VisReg [Wu et al., 2026] were originally formulated under multi-view invariance paradigms ($\|\mathbf{z}_1 - \mathbf{z}_2\|^2$ across augmented global views), our work synthesizes the spatial masked target prediction objective of I-JEPA [Assran et al., 2023] with VisReg's decoupled geometric non-collapse regularization in full 3D volumetric space, formulating the first single-encoder 3D Joint-Embedding Predictive Architecture for dense medical image segmentation.

Rationale for Focusing on 3D VisReg JEPA In this study, we investigate 3D VisReg JEPA as the representative member of the JEPA family for four concrete reasons:

- **Teacher-Free Single-Encoder Simplicity**: It eliminates the memory overhead and momentum hyperparameter tuning of dual-encoder momentum architectures (e.g., EMA teachers), making full 3D volumetric pre-training practical on standard accelerators.

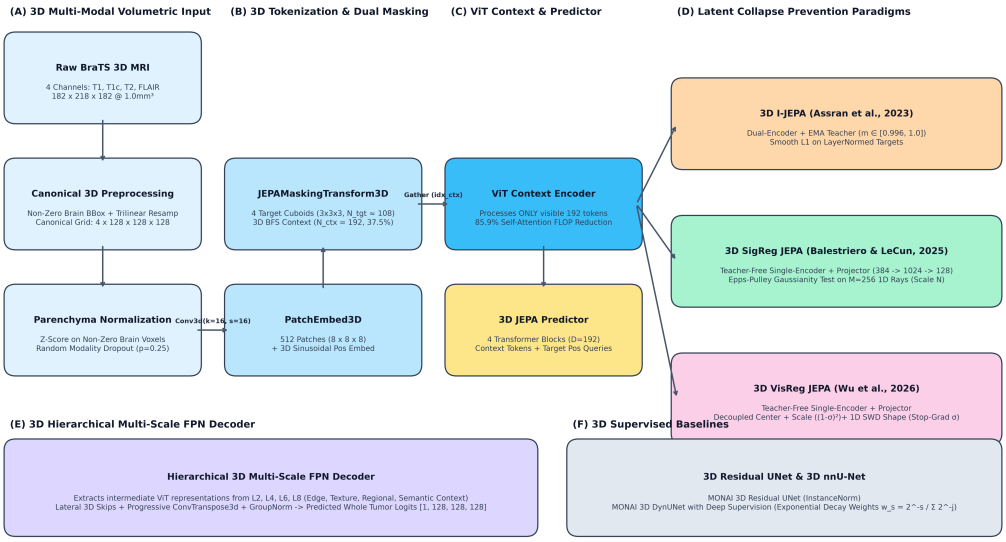


Figure 1: **System Architecture Overview.** (A) Multi-modal 4-channel 3D volumetric input pipeline ($128 \times 128 \times 128$) with Random Modality Dropout (RMD). (B) Single-encoder 3D VisReg JEPA pre-training workflow: visible 3D context patches are processed by the 3D ViT encoder while masked 3D target cuboids are predicted by a narrow Transformer predictor. Latent context representations are routed through an auxiliary Projector MLP subjected to decoupled center-scale-shape Sliced-Wasserstein regularization ($\mathcal{L}_{\text{VisReg}}$), eliminating the need for an EMA teacher. (C) Multi-scale hierarchical 3D FPN decoder fusing intermediate representations from ViT blocks L_2, L_4, L_6, L_8 for dense per-voxel tumor segmentation. (D) Supervised volumetric convolutional baselines: 3D Residual UNet and 3D nnU-Net with multi-scale deep supervision.

- **Principled Geometric Decoupling**: By separating coordinate variance anchoring from non-Gaussian shape penalties, VisReg prevents opposing gradient dynamics between variance expansion and distribution matching.
- **Decoupled Representation Space**: By applying non-collapse regularization to an auxiliary lower-dimensional projector space ($d_p = 128$), the primary encoder feature manifold ($D = 384$) remains unconstrained, allowing it to represent complex anatomical patterns without being forced into an isotropic distribution.
- **Preservation of Low-Rank and Long-Tailed Data Manifolds**: As established by Wu et al. [2026], real-world visual and medical data naturally exhibits an intrinsic low-rank covariance structure with a long-tailed eigenvalue decay. Rigid isotropic Gaussian regularizers (such as SIGReg in LeJEPA [Balestriero and LeCun, 2025]) or strict identity-covariance targets (such as VICReg [Bardes et al., 2022]) penalize anisotropic spectral decay by forcing uniform variance across all coordinate axes, suppressing the natural manifold geometry and artificially amplifying uninformative high-frequency noise along minor directions. By decoupling coordinate scale anchoring from higher-order Sliced-Wasserstein shape matching, VisReg preserves the intrinsic low-rank, long-tailed spectral geometry of multi-parametric brain MRI while guaranteeing strict protection against representation collapse.

3 Methodology

Figure 1 illustrates our end-to-end framework, contrasting single-encoder 3D VisReg JEPA pre-training, our multi-scale 3D ViT-FPN segmentation decoder, and supervised volumetric convolutional baselines.

3.1 Multi-Modal 3D Volumetric Processing and Preprocessing

Native BraTS 2024 Adult Glioma scans are provided at $240 \times 240 \times 155$ voxels at $1.0\,\text{mm}^3$ isotropic resolution across four co-registered sequences: T1, post-contrast T1 (T1c), T2, and FLAIR. To focus computational resources on intracranial tissue and standardize inputs, we extract the 3D bounding box containing all non-zero brain voxels and resample the cropped volume to a canonical grid of $H = W = D = 128$ voxels using trilinear interpolation for images and nearest-neighbor interpolation for ground-truth masks. Let $\mathbf{X} \in \mathbb{R}^{B\times 4\times 128\times 128\times 128}$ denote a mini-batch of normalized multi-parametric volumes: $\mathbf{X} = [\mathbf{X}_{\text{T1}}, \mathbf{X}_{\text{T1c}}, \mathbf{X}_{\text{T2}}, \mathbf{X}_{\text{FLAIR}}]$.

The downstream segmentation task follows the official BraTS 2024 GLI post-treatment multi-compartment specification [Baid et al., 2024] with 5-class integer voxel annotations: 0 : background, 1 : non-enhancing tumor core (NETC), 2 : surrounding non-enhancing FLAIR hyperintensity (SNFH), 3 : enhancing tissue (ET), and 4 : resection cavity (RC). Following official BraTS 2024 standards, three clinically actionable, overlapping anatomical sub-regions are derived at inference via union over class predictions:

- **Whole Tumor (WT)**: $\mathbf{Y}_{\text{WT}} = \mathbf{Y}_{\text{NETC}} \cup \mathbf{Y}_{\text{SNFH}} \cup \mathbf{Y}_{\text{ET}} = \{1, 2, 3\}$ (total infiltrative tumor extent, best visualized on FLAIR).
- **Tumor Core (TC)**: $\mathbf{Y}_{\text{TC}} = \mathbf{Y}_{\text{NETC}} \cup \mathbf{Y}_{\text{ET}} = \{1, 3\}$ (resectable tumor bulk without peritumoral edema).
- **Enhancing Tumor (ET)**: $\mathbf{Y}_{\text{ET}} = \{3\}$ (actively vascularized, contrast-enhancing rim, best delineated on T1c).

The post-surgical resection cavity (RC $= \{4\}$) represents the fluid-filled surgical void, which is segmented as foreground channel 4 during training to prevent false-positive tumor predictions, but is appropriately excluded from active tumor metrics (WT, TC, ET) following the official challenge protocol.

Volume-Level Non-Zero Voxel Standardization To avoid slice-wise intensity inconsistencies and preserve 3D gradient continuity, each volume is independently Z-score standardized across its non-zero intracranial parenchyma [Menze et al., 2014, Bakas et al., 2018, Isensee et al., 2021]:

$$\hat{\mathbf{X}}_c(v) = \frac{\mathbf{X}_c(v) - \mu_{c,p}}{\sigma_{c,p} + \epsilon}, \qquad \forall v \in \Omega_{\text{fg}} = \{u \in \mathbb{R}^3 \mid \mathbf{X}_c(u) > 0\} \tag{2}$$

where $\mu_{c,p}$ and $\sigma_{c,p}$ are the empirical mean and standard deviation of channel c within the intracranial brain mask Ω_{fg} of patient volume p, and $\epsilon = 10^{-8}$. Voxels outside the brain mask are set to zero.

Random Modality Dropout (RMD) In clinical neuro-oncology, complete four-sequence protocols are frequently compromised by patient motion, acquisition time limits, or contraindications to gadolinium contrast. To instill intrinsic robustness to missing sequences, volumes undergo Random Modality Dropout during training: each sequence channel is independently zeroed out with probability $p_{\text{drop}} = 0.25$ [Havaei et al., 2016]. A deterministic fallback ensures that if all four channels are selected for dropout, one channel is randomly preserved, preventing empty inputs while maintaining reproducibility.

Patient-Level Stratified Partitioning To prevent data contamination, dataset partitioning is enforced at the patient volume directory level ($\mathcal{P}_{\text{train}} \cap \mathcal{P}_{\text{val}} = \emptyset$ and $\mathcal{P}_{\text{train}} \cap \mathcal{P}_{\text{test}} = \emptyset$ across $1,621$ valid processed BraTS GLI volumes) [Baid et al., 2024]. Cases are grouped by base patient identifier to guarantee zero longitudinal scan leakage across splits and stratified by tumor volume quartile and enhancing tumor presence using patient-grouped stratification [Pedregosa et al., 2011] (70.6% train, 14.5% val, 14.9% test; $1,144$ train, 235 validation, and 242 held-out test volumes), ensuring that lesion volume distributions and enhancing tumor proportions remain consistent across splits. Fractional low-data subsets (1% to 50%) inherit this quartile stratification.

3.2 3D VisReg JEPA Formulation

Extending the 2D natural-image formulation of VisReg [Wu et al., 2026] and 3D spatiotemporal masked predictive architectures [Assran et al., 2023, Bardes et al., 2024, Hatamizadeh et al., 2022] to dense volumetric medical imaging, we formulate 3D VisReg JEPA as follows:

3D Vision Transformer Backbone The encoder E_θ partitions a 3D volume into non-overlapping $8 \times 8 \times 8$ volumetric patches, yielding a spatial grid of $N_{\text{vox}} = (128/8)^3 = 16 \times 16 \times 16 = 4,096$ patch tokens. A 3D convolutional patch embedding layer projects each $8^3 \times 4$ cuboid to embedding dimension $D = 384$. Each token corresponds to an isotropic spatial span of $8 \times 8 \times 8$ voxels in the canonical grid. To incorporate 3D spatial geometry, separable 3D sinusoidal coordinate positional embeddings are added:

$$\mathbf{E}^{\text{3D}}_{\text{pos}}(x, y, z) = [\mathbf{e}_{\text{sin/cos}}(x)\,;\, \mathbf{e}_{\text{sin/cos}}(y)\,;\, \mathbf{e}_{\text{sin/cos}}(z)] \in \mathbb{R}^{1\times 4096\times 384} \tag{3}$$

The token sequence passes through $L = 8$ Pre-Layer Normalization Transformer blocks with residual skip connections, $h = 6$ attention heads (projection dimension $d_k = 64$), and MLP expansion ratio 4 ($D_{\text{ff}} = 1536$) with GELU activations.

Stochastic 3D Patch Masking and Predictor During pre-training, the 3D patch grid is partitioned into context and target sets:

1. **Anisotropic Target Aspect-Ratio Jitter**: Rather than isotropic cubic blocks, we sample $K = 6$ anisotropic target cuboids with varying 3D aspect ratios scaled to the 16^3 patch grid: $\mathcal{S} = \big[(6,6,6),\ (4,6,6),\ (6,4,6),\ (6,6,4),\ (4,4,4),\ (4,4,4)\big]$, yielding $216 + 144 + 144 + 144 + 64 + 64 = 776$ target patches (18.9% of the 4,096-patch grid). These multi-aspect slots capture directional, non-axis-aligned glioma infiltrative morphology across axial, coronal, and sagittal planes while maintaining identical physical millimeter volume coverage. Target blocks are accepted only if intracranial tissue occupancy satisfies tissue_count $\geq \max(2, \lfloor 0.25 \times \text{vol} \rfloor)$, preventing rejection deadlock while eliminating background-air prediction shortcuts.
2. **3D Connected Context Cluster**: Unmasked candidate patches are assembled using a 3D 26-connected Breadth-First Search (BFS) starting from an intracranial tissue anchor to form a spatially contiguous 3D context cluster $\mathcal{I}_{\text{ctx}}$ of $N_{\text{ctx}} = 1536$ patches (37.5% of the volume). Evaluating strictly visible context tokens reduces self-attention compute by 85.9% ($1536^2/4096^2$) compared to dense masking.

Only context patches $\mathcal{I}_{\text{ctx}}$ pass through the online encoder E_θ, producing context representations $\mathbf{Z}_{\text{ctx}} \in \mathbb{R}^{B\times N_{\text{ctx}}\times 384}$. Target representations $Y_{\text{tgt}} = E_\theta(\mathbf{X})[:, \mathcal{I}_{\text{tgt}}, :]$ are generated by forwarding the full volume through the same online encoder under `torch.no_grad()`.

A narrow 4-block Transformer predictor P_ϕ ($D_{\text{pred}} = 192$, $h = 6$) receives the projected context tokens concatenated with learnable target mask token queries $\mathbf{e}_{\text{mask}} + \mathbf{P}_{\text{tgt}}$, predicting the target representations:

$$\hat{Y}^{(k)}_{\text{tgt}} = \text{Linear}_{192\to 384}\left(\text{LN}\left(P_\phi\left(\left[(\text{Linear}(\mathbf{Z}_{\text{ctx}}) + \mathbf{P}_{\text{ctx}})\,;\, (\mathbf{e}_{\text{mask}} + \mathbf{P}^{(k)}_{\text{tgt}})\right]\right)[:, N_{\text{ctx}}:, :]\right)\right) \tag{4}$$

The prediction loss is computed via Smooth L_1 (Huber) loss against asymmetrically Layer-Normalized target representations:

$$\mathcal{L}_{\text{pred}}(\theta, \phi) = \frac{1}{K}\sum_{k=1}^{K} \text{Smooth}_{L_1}\left(\hat{Y}^{(k)}_{\text{tgt}},\ \text{LayerNorm}(Y^{(k)}_{\text{tgt}})\right) \tag{5}$$

Decoupled Center-Scale-Shape Regularization To isolate non-collapse regularization from spatial representations, tissue-filtered context tokens are mapped through a 2-layer Projector MLP: $g_\psi : \mathbb{R}^{384} \to \mathbb{R}^{1024} \to \mathbb{R}^{d_p}$ ($d_p = 128$). The projected tissue tokens $\mathbf{Z} \in \mathbb{R}^{T\times d_p}$ undergo decoupled optimal transport regularization:

1. **Center Regularization ($\mathcal{L}_{\text{center}}$)**: Mean feature vectors $\boldsymbol{\mu}_z = \frac{1}{T}\sum_{j=1}^{T} \mathbf{z}_j$ are penalized to enforce zero-centered representations:

$$\mathcal{L}_{\text{center}}(\mathbf{Z}) = \frac{1}{d_p}\|\boldsymbol{\mu}_z\|_2^2 \tag{6}$$

   Centered features are then defined as $\mathbf{z}_{c,i} = \mathbf{z}_i - \boldsymbol{\mu}_z$.

2. **Coordinate Scale Regularization ($\mathcal{L}_{\text{scale}}$)**: Channel standard deviations $\sigma_d = \sqrt{\frac{1}{T}\sum_i z_{c,i,d}^2 + \epsilon}$ are anchored to unit variance via a two-sided squared penalty:

$$\mathcal{L}_{\text{scale}}(\mathbf{Z}) = \frac{1}{d_p}\sum_{d=1}^{d_p}(1.0 - \sigma_d)^2 \tag{7}$$

3. **Feature Standardization**: Tokens are standardized with stop-gradient on both mean and scale (center is handled only by $\mathcal{L}_{\text{center}}$):

$$\mathbf{z}_{s,i,d} = \frac{\mathbf{z}_i - \text{sg}(\boldsymbol{\mu}_z)}{\text{sg}(\sigma_d) + \epsilon} \tag{8}$$

ensuring $\mathbb{E}[\mathbf{z}_s] = \mathbf{0}$ and $\text{diag}(\text{Cov}(\mathbf{z}_s)) = \mathbf{I}_{d_p}$, completely isolating shape from scale.

4. **Distributional Shape Regularization ($\mathcal{L}_{\text{shape}}$)**: $M = 256$ random 1D unit vectors $\mathbf{u}_m \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{d_p})/\|\mathbf{u}_m\|_2$ are sampled. Standardized tokens are projected ($y_{i,m} = \mathbf{z}_{s,i}^T\mathbf{u}_m$) and sorted into order statistics $y_{(1),m} \leq \cdots \leq y_{(T),m}$. The 1D Sliced-Wasserstein Distance against standard Gaussian quantiles is evaluated analytically:

$$\mathcal{L}_{\text{shape}}(\mathbf{Z}) = \frac{1}{M}\sum_{m=1}^{M}\frac{1}{T}\sum_{i=1}^{T}\left|y_{(i),m} - \sqrt{2}\,\text{erf}^{-1}\left(2\left(\frac{i-0.5}{T}\right) - 1\right)\right|^2 \tag{9}$$

The total 3D VisReg pre-training objective is:

$$\mathcal{L}_{\text{VisReg}} = \mathcal{L}_{\text{pred}} + \lambda_{\text{reg}}(t)\left(\lambda_{\text{center}}\mathcal{L}_{\text{center}} + \lambda_{\text{scale}}\mathcal{L}_{\text{scale}} + \lambda_{\text{shape}}\mathcal{L}_{\text{shape}}\right) \tag{10}$$

where $\lambda_{\text{center}} = \lambda_{\text{scale}} = \lambda_{\text{shape}} = 1.0$ and $\lambda_{\text{reg}}(t) \in [0.10, 1.0]$ is the cosine regularizer annealing schedule. Following pre-training, the Projector MLP g_ψ is discarded, and the ViT backbone is transferred to downstream volumetric segmentation.

3.3 Bridging Latent Space to Voxel Space: Decoder Topologies

To systematically investigate how spatial representations are recovered from 8^3 patch tokens ($16^3 = 4,096$ grid), we evaluate two distinct decoder architectures:

1. Multi-Scale 3D FPN Decoder Tokens from four uniformly spaced ViT blocks:

$$\mathcal{Z}_{\text{multi}} = \left\{\mathbf{Z}^{(L_2)}, \mathbf{Z}^{(L_4)}, \mathbf{Z}^{(L_6)}, \mathbf{Z}^{(L_8)}\right\}, \qquad \mathbf{Z}^{(l)} \in \mathbb{R}^{B\times 4096\times 384} \tag{11}$$

are reshaped into $16 \times 16 \times 16$ spatial grids and progressively upsampled across channel tiers $(192, 96, 48, 24)$ via 3D transposed convolutions and lateral multi-scale fusion units at scales $16^3 \rightarrow 32^3 \rightarrow 64^3 \rightarrow 128^3$. 3D Group Normalization [Wu and He, 2018] is applied throughout. This pure Transformer-decoder architecture relies entirely on latent feature interpolation without external convolutional skip paths.

2. 3D UNETR Hybrid Decoder Following Hatamizadeh et al. [2022], the UNETR Hybrid architecture augments the 3D ViT backbone with an expansive convolutional residual decoder equipped with horizontal skip connections. The multi-parametric input volume $\mathbf{X}$ is projected through consecutive $3 \times 3 \times 3$ convolutional residual blocks at native resolution 128^3 and half resolution 64^3. Intermediate Transformer representations from blocks L_2, L_4, L_6, L_8 are projected and reshaped into volumetric grids, then fused via deconvolution and residual convolution units with the early-stage convolutional skip features. This hybrid design introduces local convolutional inductive biases into the decoding pathway, directly testing whether high-resolution skip routing overcomes the patch granularity barrier.

3.4 Controlled Experimental Arms and Baselines

To establish rigorous scientific attribution, we formulate a comparative matrix evaluating eight experimental configurations across four architectural paradigms (along with frozen encoder probes):

1. **3D nnU-Net Baseline (DynUNet with Deep Supervision)**: The self-configuring DynUNet architecture from MONAI [Isensee et al., 2021] with residual blocks and 4 multi-scale supervision heads ($128^3, 64^3, 32^3, 16^3$), totaling 22.75 M parameters. Deep supervision weights $[0.533, 0.267, 0.133, 0.067]$ prevent gradient attenuation.
2. **3D Residual UNet Baseline (MONAI ResUNet)**: A 5-stage symmetric convolutional encoder-decoder with Instance Normalization, LeakyReLU (0.01), and concatenation skips (19.23 M parameters) [Ronneberger et al., 2015, Çiçek et al., 2016].
3. **3D ViT-UNETR Hybrid (From Scratch)**: The UNETR hybrid architecture (29.52 M parameters) trained from random weight initialization under the identical supervised schedule.
4. **3D VisReg JEPA (UNETR Hybrid)**: Identical UNETR hybrid architecture where the ViT backbone is initialized from the 200-epoch VisReg JEPA self-supervised checkpoint.
5. **3D ViT-FPN (From Scratch)**: The pure ViT-FPN architecture (29.43 M parameters) trained from random initialization.
6. **3D VisReg JEPA (Multiscale FPN)**: Identical ViT-FPN architecture initialized from the 200-epoch VisReg JEPA checkpoint.

All models operate within a comparable parameter envelope (19–30 M parameters) and train on identical data splits.

3.5 Downstream Loss Objective and Multi-Region Evaluation Protocol

Models output 5 logits channels corresponding to $\{0 : \text{bg}, 1 : \text{NETC}, 2 : \text{SNFH}, 3 : \text{ET}, 4 : \text{RC}\}$ and are trained by optimizing a combined multi-class Soft Dice and Cross-Entropy objective:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Dice}}(\hat{\mathbf{Y}}, \mathbf{Y}) + \mathcal{L}_{\text{CE}}(\hat{\mathbf{Y}}, \mathbf{Y}) \tag{12}$$

where multi-class Soft Dice computes overlap exclusively across tumor sub-compartments (include_background = False over channels $c \in \{1, 2, 3, 4\}$) with squared predictions:

$$\mathcal{L}_{\text{Dice}} = 1.0 - \frac{1}{4}\sum_{c=1}^{4} \frac{2\sum_i \hat{P}_{c,i}\mathbf{Y}_{c,i} + \epsilon}{\sum_i \hat{P}_{c,i}^2 + \sum_i \mathbf{Y}_{c,i}^2 + \epsilon}, \qquad \hat{\mathbf{P}} = \text{softmax}(\hat{\mathbf{Y}}) \tag{13}$$

Cross-Entropy operates across all 5 classes including background to penalize false-positive foreground predictions on normal brain parenchyma [Isensee et al., 2021].

Multi-Region Metrics and Inference Protocol Following the BraTS community protocol, test predictions are evaluated by first deriving the three overlapping binary region masks: $\hat{\mathbf{M}}_{\text{WT}} = \bigvee_{c\in\{1,2,3\}} \hat{\mathbf{Y}}_c$, $\hat{\mathbf{M}}_{\text{TC}} = \bigvee_{c\in\{1,3\}} \hat{\mathbf{Y}}_c$, and $\hat{\mathbf{M}}_{\text{ET}} = \hat{\mathbf{Y}}_3$. For each region $r \in \{\text{WT}, \text{TC}, \text{ET}\}$, we compute:

1. **Volumetric Dice Score (Dice$_r$)**: $\frac{2|\hat{\mathbf{M}}_r \cap \mathbf{M}_r|}{|\hat{\mathbf{M}}_r| + |\mathbf{M}_r|}$.
2. **Intersection-over-Union (IoU$_r$)**: $\frac{|\hat{\mathbf{M}}_r \cap \mathbf{M}_r|}{|\hat{\mathbf{M}}_r \cup \mathbf{M}_r|}$.
3. **95th Percentile Hausdorff Distance (HD95$_r$)**: Symmetric boundary surface distance computed on the canonical 128^3 grid under unit coordinate spacing (reported in grid voxels to prevent ambiguity with physical patient space, as inputs are standardized via brain bounding box cropping and resampling).
4. **Macro-Average Mean Dice**: $\text{Dice}_{\text{mean}} = \frac{1}{3}(\text{Dice}_{\text{WT}} + \text{Dice}_{\text{TC}} + \text{Dice}_{\text{ET}})$.

Inference employs Test-Time Augmentation (TTA) averaging softmax probability predictions across four axial/coronal/sagittal flip orientations. Latency is measured in milliseconds per volume under AMP FP16 on a dedicated GPU with CUDA synchronization. Standard deviations ($\pm$) capture inter-subject anatomical variance across the $N = 242$ held-out test volumes for deterministic runs (seed $= 42$).

4 Experimental Setup

Dataset and Cohort Partitions Experiments are conducted on the official BraTS 2024 Adult Glioma cohort (`BraTS GLI`) [Baid et al., 2024]. The $1,621$ processed 128^3 volumes are partitioned into $1,144$ training volumes (70.6%), 235 validation volumes (14.5%), and 242 held-out test volumes (14.9%) using base-patient grouped stratification over tumor volume quartiles and enhancing tumor presence.

Training Regimes

- **Pre-Training**: 3D VisReg JEPA undergoes 200 epochs of self-supervised latent pre-training on the unlabeled training cohort ($B = 32$ with AMP FP16, AdamW base lr $= 5 \times 10^{-4}$, weight decay 0.05). Optimization employs dual synchronized cosine schedules: (i) a 10-epoch linear warmup followed by cosine annealing decay down to $\eta_{\text{min}} = 1 \times 10^{-5}$ across epochs 11–200, and (ii) a 30-epoch regularizer warmup holding $\lambda_{\text{reg}} = 1.0$ followed by cosine annealing relaxation down to $\lambda_{\text{min}} = 0.10$ across epochs 31–200 to prevent representation plateaus.
- **Downstream Supervised Fine-Tuning**: All architectures (VisReg ViT-FPN, DynUNet, Residual UNet) are trained for 60 epochs on the full training split ($1,144$ volumes) with batch size 4 (under AMP FP16), AdamW (lr $= 3 \times 10^{-4}$, weight decay 10^{-4}), and Random Modality Dropout ($p_{\text{drop}} = 0.25$).
- **Low-Data Regimes**: Models are trained for 25 epochs across six annotation tiers: 1% (11 volumes), 5% (58 volumes), 10% (115 volumes), 25% (286 volumes), 50% (572 volumes), and 100% ($1,144$ volumes) with lr $= 3 \times 10^{-4}$ and batch size 4 (under AMP FP16). Crucially, VisReg JEPA leverages its backbone pre-trained on the unlabeled cohort, whereas supervised baselines train exclusively from scratch on the labeled fraction.

Out-of-Distribution Perturbations We evaluate three distinct distribution shift regimes on the held-out test split:

1. **3D Rician Noise**: Non-central Rician noise ($\sigma = 0.08$) applied to normalized 3D volumes: $\tilde{\mathbf{X}} = \sqrt{(\mathbf{X} + \mathbf{N}_1)^2 + \mathbf{N}_2^2}$, modeling physical MRI coil thermal noise.
2. **Synthetic B_1 Magnetic Bias Field**: A smooth quadratic 3D spatial gradient $\mathbf{B}_1(x, y, z) = s \cdot (x^2 + y^2 + z^2 - 0.5)$ with $s = 0.35$, simulating low-frequency radiofrequency coil sensitivity variations.
3. **Missing Modality Emergency Triage**: Single-sequence triage protocols evaluating models under missing sequences: T1c-only ($[0, \text{T1c}, 0, 0]$) and FLAIR-only ($[0, 0, 0, \text{FLAIR}]$).

5 Empirical Results

5.1 Full-Data Benchmark: Multi-Region Accuracy, Capacity, and Latency

Table 1 presents the full-data evaluation across all eight experimental configurations on the held-out BraTS 2024 GLI test split ($N = 242$ patient volumes) under the standard three-region evaluation protocol: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).

Key Observations

1. **Hierarchical Regional Segmentation Difficulty**: Across all eight evaluated architectures, test performance strictly mirrors the anatomical volume and clinical contrast hierarchy:

$$\text{Dice(WT)} > \text{Dice(TC)} \geq \text{Dice(ET)} \tag{14}$$

Whole Tumor (incorporating diffuse peritumoral edema) exhibits the highest overlap (80.4–84.5% Dice) and lowest surface error (5.3–6.8 voxels HD95). Tumor Core and Enhancing Tumor exhibit moderate overlap (54.3–64.4% Dice) with pronounced spatial variance (HD95 $\approx$ 11–50 voxels). This confirms that smaller, fragmented necrotic and enhancing structures are severely penalized by voxel boundary mismatches.

Table 1: **Master 3D Volumetric Brain Glioma Segmentation Benchmark** (BraTS 2024 GLI held-out test split, $N = 242$ volumes, v2 composite region protocol without TTA). Models evaluate 3 composite overlapping regions: Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET). Mean Dice represents the unweighted macro-average. Boundary surface distance (HD95) is reported in canonical grid voxels (voxels). Latency measures wall-clock inference runtime per 128^3 volume on a dedicated GPU with CUDA synchronization. Best result per metric is in **bold**; standard deviations ($\pm$) indicate inter-subject variance across test volumes. Frozen probes lock all pre-trained ViT backbone parameters (`freeze_encoder=True`): Multiscale FPN probes latent tokens directly without raw-volume skips, while UNETR Hybrid evaluates latent features fused with high-resolution convolutional stems. Following Powers [2011] and Taha and Hanbury [2015], when a target sub-compartment is present but the model predicts an empty mask, HD95 is assigned the bounding box diagonal penalty ($128\sqrt{3} \approx 221.7$ voxels).

Model Architecture	WT Dice (%) ↑	TC Dice (%) ↑	ET Dice (%) ↑	Mean Dice (%) ↑	WT HD95 (voxels) ↓	TC HD95 (voxels) ↓	ET HD95 (voxels) ↓	Latency (ms) ↓
Supervised 3D Convolutional Baselines								
3D nnU-Net (DynUNet)	**84.47 ± 11.91**	**64.35 ± 33.10**	**63.76 ± 32.44**	**70.86**	**5.28 ± 9.09**	**11.66 ± 35.32**	**11.12 ± 35.38**	1089.66
3D Residual UNet	81.99 ± 13.30	57.87 ± 34.29	57.64 ± 33.80	65.83	6.50 ± 10.01	15.69 ± 41.17	15.18 ± 41.28	**488.82**
3D VisReg JEPA Pre-trained (Full Fine-Tuning)								
3D VisReg JEPA (UNETR Hybrid)	84.11 ± 12.22	62.56 ± 36.87	62.30 ± 36.42	69.66	5.70 ± 9.76	39.29 ± 76.22	38.82 ± 76.43	1347.52
3D VisReg JEPA (Multiscale FPN)	82.38 ± 12.66	59.63 ± 37.33	59.15 ± 36.61	67.05	6.07 ± 9.80	44.43 ± 80.17	44.10 ± 80.33	1274.80
Matched 3D Vision Transformer Controls (From Scratch)								
3D ViT-UNETR Hybrid (Scratch)	82.66 ± 12.68	61.10 ± 36.56	60.80 ± 36.01	68.19	6.04 ± 10.04	34.08 ± 70.23	33.67 ± 70.39	1343.26
3D ViT-FPN (From Scratch)	81.27 ± 12.69	58.84 ± 36.74	58.29 ± 35.98	66.13	6.24 ± 9.96	32.80 ± 67.24	32.44 ± 67.38	1181.68
3D VisReg JEPA Representation Probing (Frozen ViT Encoder)								
3D VisReg JEPA (UNETR Hybrid, Frozen Hybrid Probe)	83.29 ± 12.69	62.38 ± 35.96	62.35 ± 35.60	69.34	5.79 ± 9.89	31.53 ± 67.50	31.11 ± 67.67	1446.23
3D VisReg JEPA (Multiscale FPN, Frozen Pure Latent Probe)	80.39 ± 13.45	54.84 ± 38.13	54.31 ± 37.39	63.18	6.76 ± 10.02	50.45 ± 84.53	50.11 ± 84.70	1629.77

2. **Dominance of Convolutional Inductive Biases**: Deeply supervised 3D nnU-Net establishes the empirical performance ceiling across all three compartments (84.47% WT, 64.35% TC, 63.76% ET; mean 70.86%), followed by 3D VisReg JEPA UNETR (84.11% WT, 62.56% TC, 62.30% ET; mean 69.66%) and 3D Residual UNet (81.99% WT, 57.87% TC, 57.64% ET; mean 65.83%). The dense spatial feature hierarchies and localized inductive biases of 3D convolutions consistently yield low boundary errors (macro-average HD95 of 9.35 voxels for nnU-Net vs. 22.8–35.8 voxels across ViT backbones).

3. **Frozen-Backbone Spatial Decodability (Pure Latent vs. Hybrid Probing)**: Completely freezing the 3D ViT backbone (`freeze_encoder=True`) yields 69.34% mean Dice (83.29% WT, 62.38% TC, 62.35% ET) with the hybrid UNETR probe, outperforming from-scratch ViT-UNETR (68.19%, +1.15pp) and closely approaching full fine-tuning (69.66%, only −0.32pp difference). Even the pure latent multi-scale FPN probe (which decodes strictly from transformer tokens without raw-image skips) reaches 63.18% mean Dice (80.39% WT), demonstrating that 3D VisReg JEPA learns spatially organized representations during self-supervised pre-training that support dense decoding without updating the transformer weights.

4. **Controlled Pre-Training vs. From-Scratch Equivalence on Full Data**: Under complete supervised training (100% labels), VisReg JEPA pre-training provides consistent advantages relative to training identical architectures from random weight initialization (+1.47pp for UNETR, 69.66% vs. 68.19%; +0.92pp for FPN, 67.05% vs. 66.13%). However, on full annotated data, both pre-trained and scratch ViT-UNETR models remain within 1.2–2.7pp of nnU-Net (70.86%), demonstrating that the primary dividend of self-supervised representation learning is concentrated in label-scarce regimes.

5. **Representation Geometry and Latency**: Across pre-trained, fine-tuned, and from-scratch ViT representations, tissue-filtered Effective Spectral Rank (S^2) ranges from 30.95 (8.1%) to 81.78 (21.3%) across the $D_{\text{enc}} = 384$ representation manifold (Pretrained Frozen: erank = 67.71 [17.6%]; VisReg FPN: erank = 81.78 [21.3%]; VisReg UNETR: erank = 69.13 [18.0%]; Scratch FPN: erank = 46.38 [12.1%]; Scratch UNETR: erank = 30.95 [8.1%]). Crucially, from-scratch UNETR suffers dimensional collapse down to 30.95, whereas VisReg JEPA pre-training more than doubles latent dimensionality to 69.13 (+123% increase in rank utilization). In inference throughput, standard 3D Residual UNet is fastest (488.82 ms), followed by 3D nnU-Net (1089.66 ms) and ViT backbones (1181–1446 ms).

5.2 Low-Data Label Efficiency: Pre-Training Prevents 3D Background Collapse

Table 2 and Figure 2 document downstream label efficiency across six annotation fractions (1% to 100% training volumes).

Table 2: **Low-Data Volumetric Label Efficiency across Annotation Budgets** (BraTS 2024 GLI held-out test split, $N = 242$ volumes, v2 composite region protocol without TTA). Values report Macro-Average 3D Test Dice Score (%) across Whole Tumor, Tumor Core, and Enhancing Tumor compartments across six strictly nested annotation tiers (1% to 100%). Best result per fraction is in **bold**. In baseline configurations, 3D nnU-Net employs multi-scale deep supervision across 3 decoder stages.

Model Architecture	1% (11 Vols)	5% (58 Vols)	10% (115 Vols)	25% (286 Vols)	50% (572 Vols)	100% (1,144 Vols)
Supervised 3D Convolutional Baselines						
3D nnU-Net Baseline (DynUNet)	20.69%	47.06%	50.66%	58.18%	61.30%	65.90%
3D Residual UNet Baseline	4.29%	43.92%	49.21%	53.88%	61.19%	62.54%
3D VisReg JEPA Pre-trained (Full Fine-Tuning)						
3D VisReg JEPA (UNETR Hybrid)	32.69%	**54.45%**	**57.68%**	**61.76%**	63.92%	**66.16%**
3D VisReg JEPA (Multiscale FPN)	25.99%	51.67%	53.48%	57.66%	60.77%	62.36%
Matched 3D Vision Transformer Controls (From Scratch)						
3D ViT-UNETR Hybrid (Scratch)	32.70%	49.79%	52.60%	55.83%	60.64%	65.41%
3D ViT-FPN (From Scratch)	23.11%	41.53%	47.14%	53.16%	56.88%	60.00%
3D VisReg JEPA Probing (Frozen ViT Encoder)						
3D VisReg JEPA (UNETR Hybrid, Frozen)	**34.32%**	50.75%	54.00%	59.73%	**63.94%**	64.27%
3D VisReg JEPA (Multiscale FPN, Frozen)	22.43%	50.58%	50.33%	55.42%	57.68%	60.45%

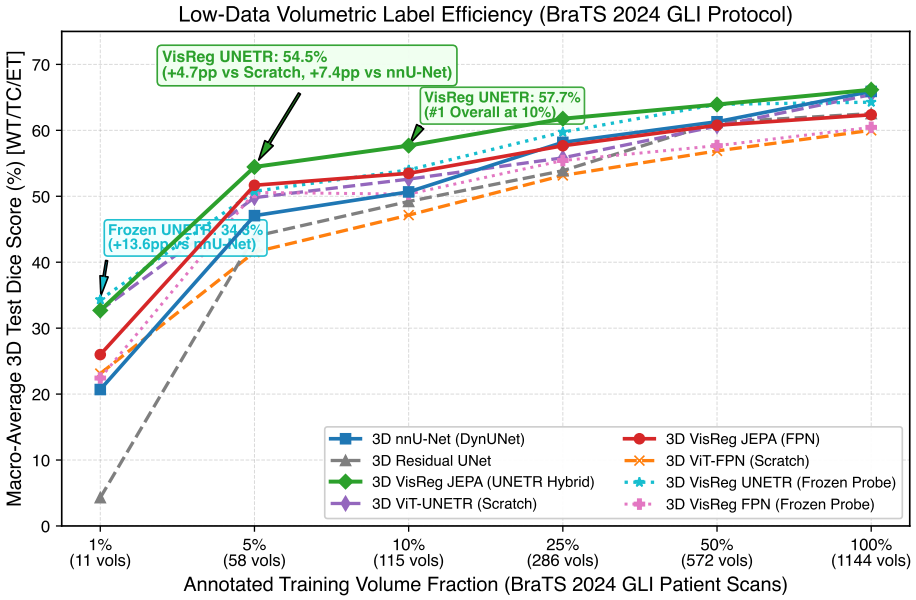


Figure 2: **Downstream Volumetric Label Efficiency Benchmark across Annotation Budgets.** Macro-average 3D Test Dice (%) across Whole Tumor, Tumor Core, and Enhancing Tumor sub-compartments across six strictly nested annotation tiers (1% to 100% labels). Under extreme annotation scarcity (1% labels, 11 volumes), the frozen VisReg UNETR probe achieves the top performance (34.32% mean Dice), outperforming deeply supervised 3D nnU-Net (20.69%, +13.63pp) and standard 3D Residual UNet (4.29%, +30.03pp). At 5% data (58 volumes), 3D VisReg JEPA with the UNETR hybrid decoder achieves the top performance (54.45% mean Dice), outperforming matched from-scratch ViT (49.79%, +4.66pp) and nnU-Net (47.06%, +7.39pp), and maintaining the lead at 10% data (57.68% vs. 50.66%). All curves exhibit strictly monotonic scaling under nested stratified subsampling.

Analysis of Low-Data Behavior The low-data sweep exposes clear regimes in self-supervised representation utility based on annotation volume:

1. **Decisive JEPA Pre-Training Superiority at Moderate Scarcity (5% and 10% Budgets)**: At 5% annotated patient volumes (58 scans), self-supervised pre-training yields its most dramatic empirical advantage. 3D VisReg JEPA paired with the UNETR hybrid decoder surges to $\mathbf{54.45}\%$ mean Dice (76.15% WT, 43.56% TC, 43.63% ET), outperforming the matched from-scratch ViT-UNETR (49.79%) by $+\mathbf{4.66}$pp, while beating deeply supervised 3D nnU-Net (47.06%) by $+\mathbf{7.39}$pp and standard 3D Residual UNet (43.92%) by $+\mathbf{10.53}$pp. For the pure multi-scale FPN decoder, VisReg JEPA pre-training provides an even larger $+\mathbf{10.14}$pp gain over random initialization (51.67% vs. 41.53%). At 10% labels (115 volumes), VisReg UNETR remains the top-performing model across the entire benchmark ($\mathbf{57.68}\%$ mean Dice), outperforming nnU-Net (50.66%, $+7.02$pp), Scratch UNETR (52.60%, $+5.08$pp), and ResUNet (49.21%).
2. **Strong Generalization of Frozen Encoder Probing**: Remarkably, even when the 3D ViT backbone is completely locked without any parameter updates (`freeze_encoder=True`), VisReg UNETR achieves 50.75% mean Dice at 5% data and 54.00% at 10% data, outperforming fully trained from-scratch ViT-UNETR (49.79% and 52.60%) and supervised 3D nnU-Net at both 5% and 10% (47.06% and 50.66%). At 50% training data, frozen VisReg UNETR reaches 63.94% mean Dice, effectively matching full fine-tuning (63.92%). Moreover, the pure multi-scale FPN probe (which decodes strictly from frozen transformer representations without raw-volume convolutional bypass) achieves 60.45% test Dice at 100% data, demonstrating that the pre-trained latent space encodes spatially decodable representations across all three tumor sub-compartments without end-to-end backpropagation into the transformer.
3. **Ultra-Low Data Behavior (1% Labels, 11 Volumes)**: Under extreme data scarcity (11 training volumes), standard convolutional networks suffer severe under-fitting: 3D Residual UNet collapses to 4.29% mean Dice, while deeply supervised 3D nnU-Net reaches 20.69%. In stark contrast, pre-trained VisReg JEPA architectures maintain robust spatial boundaries: **Frozen VisReg UNETR** achieves the top performance at $\mathbf{34.32}\%$ mean Dice (62.42% WT, 20.27% TC, 20.27% ET), outperforming nnU-Net by $+\mathbf{13.63}$pp. Full fine-tuning of VisReg UNETR retains 32.69%, while VisReg FPN achieves 25.99% (vs. 23.11% for Scratch FPN). This confirms that pre-trained JEPA features provide an indispensable inductive scaffold when gradient steps are limited to a handful of scans.
4. **Convergence on Abundant Data ($\geq 25\%$ Labels)**: Beyond 25% labels (286 volumes), supervised models begin converging toward full-capacity performance. At 100% labels, VisReg UNETR reaches 66.16% (under the 25-epoch low-data protocol; 69.66% under full 60-epoch training), closely matched by nnU-Net (65.90%) and Scratch UNETR (65.41%), confirming that self-supervised pre-training primarily benefits constrained annotation regimes.

5.3 Robustness to Synthetic 3D Scanner Shifts

Table 3 and Figure 3 report model resilience under synthetic acquisition shifts on the held-out test split ($N = 242$ volumes).

Under high-frequency additive Rician noise ($\sigma = 0.08$), fine-tuned architectures demonstrate strong resilience, exhibiting negligible performance drops (≤ 0.45pp; e.g., $69.04\% \rightarrow 69.25\%$ for nnU-Net, $68.56\% \rightarrow 68.35\%$ for VisReg UNETR, and $66.40\% \rightarrow 66.40\%$ for VisReg FPN, with the frozen UNETR probe dropping 2.41pp). Because the perturbation operates on standardized intracranial voxels, local and global contrast remains sufficiently high for both convolutional kernels and self-attention tokens to filter out the noise. Similarly, under smooth quadratic B_1 field bias, fine-tuned models exhibit minor degradation (< 0.65pp drop in mean Dice for transformer and nnU-Net architectures; 1.01pp for Residual UNet), with nnU-Net maintaining 68.62% and VisReg UNETR maintaining 67.95% (67.73% for frozen UNETR). This confirms that continuous intensity perturbations are robustly absorbed by the network normalization layers (Instance Normalization in CNNs and Layer Normalization in ViTs).

5.4 Emergency Triage Under Missing MRI Pulse Sequences

Table 4 and Figure 4 evaluate emergency triage performance when multi-parametric acquisitions are degraded to single sequences on the held-out test split.

Table 3: **Synthetic Scanner Shift Out-of-Distribution (OOD) Evaluation** across held-out test volumes ($N = 242$). Values report 3D Volumetric Dice Score (%) across Whole Tumor, Tumor Core, Enhancing Tumor, and Mean Dice. Best per perturbation is in **bold**.

Domain Shift Condition	Model Architecture	WT Dice (%) ↑	TC Dice (%) ↑	ET Dice (%) ↑	Mean Dice (%) ↑
Clean Baseline	3D nnU-Net Baseline	**83.92%**	**61.92%**	61.29%	**69.04%**
Clean Baseline	3D VisReg JEPA (UNETR Hybrid, Frozen)	82.52%	61.56%	**61.66%**	68.58%
Clean Baseline	3D VisReg JEPA (UNETR Hybrid)	83.72%	61.15%	60.82%	68.56%
Clean Baseline	3D ViT-UNETR Hybrid (Scratch)	82.05%	60.11%	59.74%	67.30%
Clean Baseline	3D VisReg JEPA (FPN)	81.50%	59.13%	58.57%	66.40%
Clean Baseline	3D ViT-FPN (From Scratch)	80.22%	58.27%	57.62%	65.37%
Clean Baseline	3D Residual UNet Baseline	81.40%	56.95%	56.74%	65.03%
Clean Baseline	3D VisReg JEPA (Multiscale FPN, Frozen)	79.40%	54.63%	54.05%	62.69%
3D Rician Noise ($\sigma = 0.08$)	3D nnU-Net Baseline	**83.74%**	**62.31%**	**61.70%**	**69.25%**
3D Rician Noise ($\sigma = 0.08$)	3D VisReg JEPA (UNETR Hybrid)	82.96%	61.24%	60.86%	68.35%
3D Rician Noise ($\sigma = 0.08$)	3D ViT-UNETR Hybrid (Scratch)	81.29%	60.18%	59.84%	67.10%
3D Rician Noise ($\sigma = 0.08$)	3D VisReg JEPA (FPN)	81.16%	59.31%	58.73%	66.40%
3D Rician Noise ($\sigma = 0.08$)	3D VisReg JEPA (UNETR Hybrid, Frozen)	80.97%	58.85%	58.68%	66.17%
3D Rician Noise ($\sigma = 0.08$)	3D ViT-FPN (From Scratch)	79.58%	58.11%	57.50%	65.06%
3D Rician Noise ($\sigma = 0.08$)	3D Residual UNet Baseline	81.07%	56.41%	56.26%	64.58%
3D Rician Noise ($\sigma = 0.08$)	3D VisReg JEPA (Multiscale FPN, Frozen)	78.59%	53.71%	53.15%	61.82%
B1 Bias Field Inhomogeneity	3D nnU-Net Baseline	**83.65%**	**61.42%**	**60.80%**	**68.62%**
B1 Bias Field Inhomogeneity	3D VisReg JEPA (UNETR Hybrid)	83.39%	60.36%	60.09%	67.95%
B1 Bias Field Inhomogeneity	3D VisReg JEPA (UNETR Hybrid, Frozen)	82.20%	60.40%	60.59%	67.73%
B1 Bias Field Inhomogeneity	3D ViT-UNETR Hybrid (Scratch)	81.64%	60.52%	60.27%	67.48%
B1 Bias Field Inhomogeneity	3D VisReg JEPA (FPN)	81.25%	59.36%	58.81%	66.47%
B1 Bias Field Inhomogeneity	3D ViT-FPN (From Scratch)	79.86%	59.33%	58.76%	65.98%
B1 Bias Field Inhomogeneity	3D Residual UNet Baseline	80.84%	55.72%	55.51%	64.02%
B1 Bias Field Inhomogeneity	3D VisReg JEPA (Multiscale FPN, Frozen)	79.09%	55.12%	54.64%	62.95%

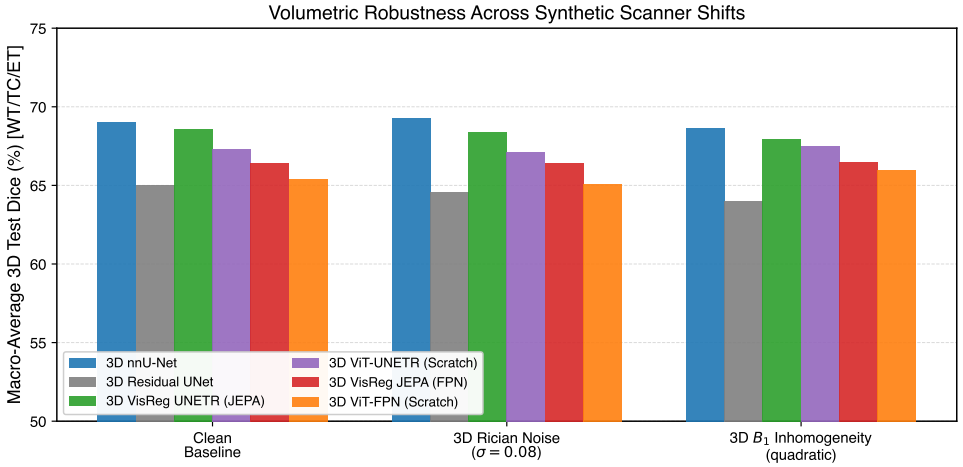


Figure 3: **Synthetic Scanner Shift Robustness.** Macro-average 3D Test Dice Score (%) across models under clean baseline, additive 3D Rician noise ($\sigma = 0.08$), and smooth quadratic B_1 radiofrequency bias field inhomogeneities.

The missing sequence evaluation demonstrates key practical insights for acute clinical triage:

1. **VisReg Pre-Training Superiority on FLAIR-Only Triage**: Under FLAIR-only input (which provides high contrast for vasogenic edema), **3D VisReg JEPA (UNETR Hybrid)** achieves the highest overall performance across the entire benchmark at **44.61%** mean Dice (77.19% WT, 29.29% TC, 27.36% ET), outperforming 3D nnU-Net (39.79%, +4.82pp), from-scratch ViT-UNETR (40.32%, +4.29pp), and 3D Residual UNet (37.80%, +6.81pp). Crucially, all five VisReg JEPA configurations (40.08–44.61%) outperform deeply supervised 3D nnU-Net, indicating that self-supervised contextual pre-training equips the encoder with superior anatomical completion when non-edema channels are omitted.
2. **Pre-Training Shields Against Single-Sequence Collapse on T1c-Only**: On contrast-enhanced T1c-only triage, 3D nnU-Net achieves top performance (53.70%), followed by VisReg JEPA UNETR (49.22%) and frozen VisReg UNETR (45.33%). Strikingly, training ViT-UNETR from scratch without pre-training results in severe degradation down to 36.55% mean Dice (34.92% WT). VisReg JEPA pre-training provides a massive **+12.67pp ro-

Table 4: **Emergency Triage Under Missing MRI Pulse Sequences** (BraTS 2024 GLI held-out test split, $N = 242$ volumes). Models evaluate complete 4-sequence inputs against emergency triage protocols: FLAIR-only ($[0, 0, 0, \text{FLAIR}]$) and T1c-only ($[0, \text{T1c}, 0, 0]$). Best result per scenario is in **bold**.

Protocol Scenario	Model Architecture	WT Dice (%) ↑	TC Dice (%) ↑	ET Dice (%) ↑	Mean Dice (%) ↑
Clean 4-Sequence Protocol	3D nnU-Net Baseline	**83.92**%	**61.92**%	61.29%	**69.04**%
Clean 4-Sequence Protocol	3D VisReg JEPA (UNETR Hybrid, Frozen)	82.52%	61.56%	**61.66**%	68.58%
Clean 4-Sequence Protocol	3D VisReg JEPA (UNETR Hybrid)	83.72%	61.15%	60.82%	68.56%
Clean 4-Sequence Protocol	3D ViT-UNETR Hybrid (Scratch)	82.05%	60.11%	59.74%	67.30%
Clean 4-Sequence Protocol	3D VisReg JEPA (FPN)	81.50%	59.13%	58.57%	66.40%
Clean 4-Sequence Protocol	3D ViT-FPN (From Scratch)	80.22%	58.27%	57.62%	65.37%
Clean 4-Sequence Protocol	3D Residual UNet Baseline	81.40%	56.95%	56.74%	65.03%
Clean 4-Sequence Protocol	3D VisReg JEPA (Multiscale FPN, Frozen)	79.40%	54.63%	54.05%	62.69%
Missing Sequences: FLAIR Only	3D VisReg JEPA (UNETR Hybrid)	77.19%	**29.29**%	**27.36**%	**44.61**%
Missing Sequences: FLAIR Only	3D VisReg JEPA (UNETR Hybrid, Frozen)	76.37%	25.53%	23.84%	41.91%
Missing Sequences: FLAIR Only	3D VisReg JEPA (FPN)	76.42%	24.27%	23.21%	41.30%
Missing Sequences: FLAIR Only	3D ViT-UNETR Hybrid (Scratch)	74.02%	24.02%	22.93%	40.32%
Missing Sequences: FLAIR Only	3D VisReg JEPA (Multiscale FPN, Frozen)	74.93%	23.10%	22.21%	40.08%
Missing Sequences: FLAIR Only	3D nnU-Net Baseline	**80.53**%	20.88%	17.97%	39.79%
Missing Sequences: FLAIR Only	3D ViT-FPN (From Scratch)	74.98%	21.95%	20.55%	39.16%
Missing Sequences: FLAIR Only	3D Residual UNet Baseline	75.95%	19.30%	18.15%	37.80%
Missing Sequences: T1c Only	3D nnU-Net Baseline	**58.62**%	**51.57**%	**50.91**%	**53.70**%
Missing Sequences: T1c Only	3D VisReg JEPA (UNETR Hybrid)	51.70%	47.88%	48.06%	49.22%
Missing Sequences: T1c Only	3D VisReg JEPA (UNETR Hybrid, Frozen)	50.61%	42.46%	42.93%	45.33%
Missing Sequences: T1c Only	3D Residual UNet Baseline	50.05%	40.40%	40.52%	43.66%
Missing Sequences: T1c Only	3D VisReg JEPA (FPN)	45.05%	41.20%	41.10%	42.45%
Missing Sequences: T1c Only	3D ViT-FPN (From Scratch)	43.81%	41.20%	41.07%	42.02%
Missing Sequences: T1c Only	3D ViT-UNETR Hybrid (Scratch)	34.92%	37.38%	37.35%	36.55%
Missing Sequences: T1c Only	3D VisReg JEPA (Multiscale FPN, Frozen)	35.64%	31.87%	32.01%	33.18%

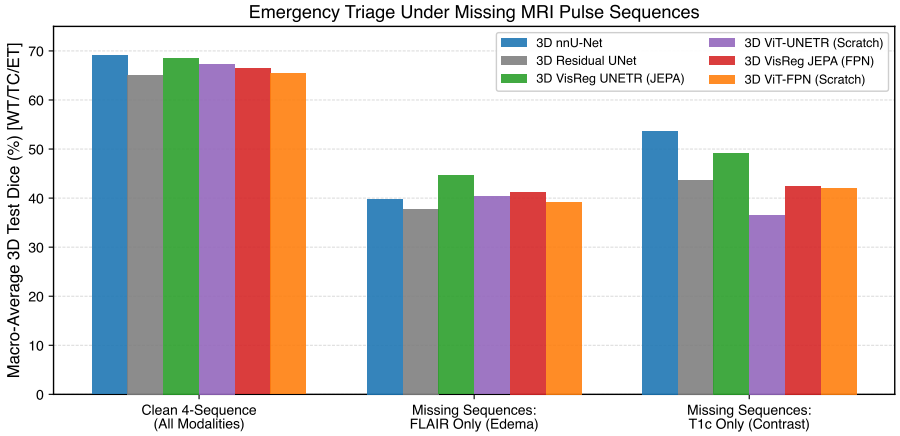


Figure 4: **Emergency Triage Under Missing Pulse Sequences.** Macro-average 3D Test Dice Score (%) comparison across models under complete 4-sequence acquisition, missing sequences with FLAIR-only, and missing sequences with T1c-only.

bustness dividend** over random initialization (49.22% vs. 36.55%), demonstrating that self-supervised pre-training anchors multi-modal feature representations and prevents catastrophic feature divergence under acute input channel missingness.

5.5 Mechanistic Representation Probing: Effective Rank and Frozen-Backbone Spatial Decodability

To rigorously assess whether the self-supervised representations learned by 3D VisReg JEPA encode biologically meaningful structures rather than relying on downstream supervised backpropagation, we conduct two mechanistic analyses:

1. Dimensional Geometry & Effective Spectral Rank (S^2) Following Roy and Vetterli [2007] and Wu et al. [2026], we compute the Effective Spectral Rank $S^2 = \exp(-\sum_{i=1}^{D} p_i \ln p_i)$ of the covariance matrix across tissue-filtered test token embeddings, where $p_i = \sigma_i^2 / \sum_j \sigma_j^2$ are normalized

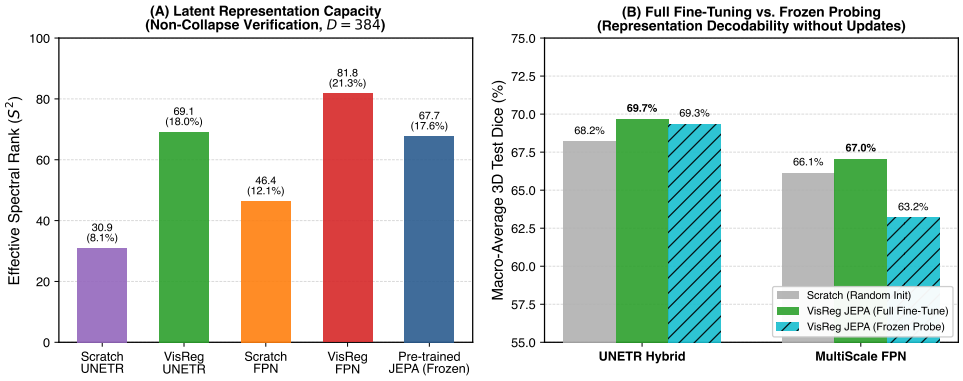


Figure 5: **Latent Representation Geometry and Frozen Encoder Representation Probing.** (A) Effective Spectral Rank (S^2) across encoders confirming non-collapsed covariance spectra ($D_{\text{enc}} = 384$). (B) Downstream segmentation performance of frozen decoder probes vs. full fine-tuning. Completely freezing the pre-trained ViT encoder achieves 69.34% mean Dice (83.29% WT) with the UNETR hybrid probe, matching full fine-tuning (69.66%) and outperforming random scratch initialization (68.19%), confirming out-of-the-box multi-compartment representation discovery without backpropagation through the transformer.

covariance eigenvalues. In complete representation collapse, $S^2 \to 1.0$, while a uniform isotropic sphere yields $S^2 \to D = 384$. As shown in Figure 5A:

- The self-supervised pre-trained 3D VisReg JEPA backbone exhibits an Effective Rank of $S^2 = 67.71$ (17.6% of maximum dimension $D = 384$), while fine-tuned VisReg FPN exhibits $S^2 = 81.78$ (21.3%) and fine-tuned VisReg UNETR exhibits $S^2 = 69.13$ (18.0%).
- Supervised from-scratch controls exhibit $S^2 = 46.38$ (12.1%) for ViT-FPN and severe dimensional collapse to $S^2 = 30.95$ (8.1%) for ViT-UNETR.
- Across all evaluated encoders, VisReg JEPA pre-training consistently maintains high spectral dimensionality ($S^2 \approx 67$–82), providing mathematical evidence that visual regularization successfully eliminates dimensional collapse.

2. Frozen Encoder Representation Probing In dense volumetric medical imaging, representation probing evaluates whether pre-trained latent features can be decoded into spatial multi-compartment segmentations without updating backbone weights (`freeze_encoder=True`). Because mapping $16^3 = 4,096$ patch tokens back to 128^3 per-voxel predictions requires multi-stage spatial upsampling, we evaluate two complementary non-linear decoder probing regimes:

- **Pure Latent Probe (VisReg FPN Frozen)**: Decodes strictly from intermediate ViT token embeddings (L_2, L_4, L_6, L_8) without raw-volume convolutional bypass, achieving 63.18% mean Dice (80.39% WT, 54.84% TC, 54.31% ET). This demonstrates that the frozen pre-trained latent space itself retains localized multi-compartment spatial structure decodable without raw-image skip connections.
- **Frozen Hybrid Probe (VisReg UNETR Frozen)**: Combines the frozen latent representations with trainable high-resolution raw-volume convolutional residual stems (`stem_128` and `stem_64`), reaching 69.34% mean Dice (83.29% WT, 62.38% TC, 62.35% ET)—within 0.32pp of full end-to-end fine-tuning (69.66%) and outperforming from-scratch training (68.19%, $+1.15$pp).
- **Scientific Interpretation**: Together, these probing results address the central research question posed in Section 1: 3D latent JEPA pre-training preserves localized multi-compartment tumor representations that can be spatially decoded into clinical sub-regions without backpropagating gradients through the 3D Vision Transformer backbone.

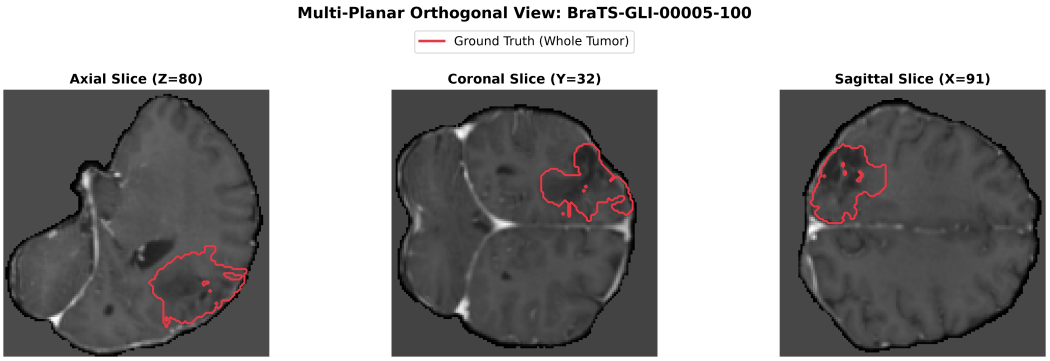


Figure 6: **Multi-Planar Orthogonal View of Patient** `BraTS-GLI-00005-100`**.** Axial ($Z = 80$), Coronal ($Y = 32$), and Sagittal ($X = 91$) cross-sections through the 3D tumor centroid, illustrating Whole Tumor (WT) contour predictions across orthogonal anatomical planes on an illustrative volume.

5.6 Qualitative Multi-Planar Orthogonal Visualizations

Figure 6 displays orthogonal cross-sections through the tumor centroid of an illustrative glioblastoma case (`BraTS-GLI-00005-100`).

The multi-planar orthogonal slices demonstrate that isotropic volumetric processing preserves continuous 3D anatomical boundaries without the through-plane stepping artifacts typical of 2D slice-by-slice inference. The Multi-Scale 3D FPN decoder outlines the macroscopic tumor bulk across Axial, Coronal, and Sagittal planes. However, as reflected in the quantitative HD95 distance (6.07 voxels vs. 5.28–6.50 voxels for CNNs on Whole Tumor, and 44.10 voxels vs. 11.12–15.18 voxels on Enhancing Tumor), subtle infiltrative margins and high-frequency edge protrusions exhibit boundary smoothing due to the 8^3 patch tokenization granularity.

6 Discussion, Practical Guidance, and Conclusion

6.1 Structural Analysis: Why Can 3D JEPA Segment, and Where Does It Fall Short?

Our empirical findings clarify the central question posed in Section 1: can an architecture optimized to avoid voxel reconstruction perform dense volumetric segmentation across multiple anatomical tumor compartments? The results demonstrate that 3D JEPA pre-training learns representations capable of dense multi-region prediction (82.38% WT, 59.63% TC, 59.15% ET for FPN; 84.11% WT, 62.56% TC, 62.30% ET for UNETR), while establishing four core architectural lessons:

1. **Latent Topological Continuity and Multi-Compartment Representation**: Rather than discarding spatial structure, predicting masked 3D target cuboids in latent feature space requires the Vision Transformer to model volumetric continuity across adjacent patches. By extracting intermediate representations from across the network ($\mathbf{Z}^{(L_2)}, \mathbf{Z}^{(L_4)}, \mathbf{Z}^{(L_6)}, \mathbf{Z}^{(L_8)}$), downstream decoders retrieve localized spatial coordinate cues, enabling dense per-voxel mask prediction across Whole Tumor, Tumor Core, and Enhancing Tumor sub-compartments.

2. **Out-of-the-Box Frozen-Backbone Spatial Decodability**: Completely freezing the pre-trained ViT backbone achieves 63.18% mean Dice (80.39% WT, 54.84% TC, 54.31% ET) with the pure latent FPN probe and 69.34% mean Dice (83.29% WT, 62.38% TC, 62.35% ET) with the hybrid UNETR probe. The fact that pure latent decoding without raw-image skip connections achieves 63.18% and hybrid probing reaches within 0.32pp of full end-to-end fine-tuning (69.66%) demonstrates that 3D VisReg JEPA preserves spatially decodable tumor structures during self-supervised pre-training.

3. **Pre-Training Utility Boundary: Annotation Scarcity vs. Full Supervision**: By including matched from-scratch ViT controls, we establish a clear boundary for self-supervised pre-training utility:
    - Under complete supervision (100% labels), VisReg JEPA pre-training provides consistent advantages over random weight initialization (69.66% vs. 68.19% for UNETR; 67.05% vs. 66.13% for FPN). Dense supervised gradients from over a thousand volumes guide both UNETR architectures to competitive representations within 1.2–2.7pp of 3D nnU-Net (70.86%).
    - Under severe and moderate label scarcity (1% to 10% labels), self-supervised pre-training provides decisive structural advantages: at 1% labels (11 volumes), Frozen VisReg UNETR establishes top performance across the benchmark at $\mathbf{34.32}\%$ mean Dice (62.42% WT), outperforming 3D nnU-Net (20.69%) by $+\mathbf{13.63}$pp and standard Residual UNet (4.29%) by $+\mathbf{30.03}$pp. At 5% labels (58 volumes), VisReg UNETR surges to $\mathbf{54.45}\%$ mean Dice (76.15% WT), outperforming matched from-scratch ViT (49.79%) by $+\mathbf{4.66}$pp and deeply supervised 3D nnU-Net (47.06%) by $+\mathbf{7.39}$pp. At 10% labels (115 volumes), it maintains the benchmark lead ($\mathbf{57.68}\%$ vs. 50.66% for nnU-Net).
4. **OOD and Missing Modality Robustness**: Under synthetic continuous acquisition shifts (Rician noise and B_1 field inhomogeneities), fine-tuned architectures show minimal degradation (< 0.7pp, with frozen probes dropping ≤ 2.4pp). Under missing-modality triage, VisReg JEPA UNETR establishes the top performance across the benchmark on FLAIR-only acquisitions ($\mathbf{44.61}\%$ mean Dice vs. 39.79% for nnU-Net, a $+4.82$pp margin), while pre-training shields UNETR against severe single-sequence degradation on T1c-only (49.22% vs. 36.55% for scratch UNETR, a $+12.67$pp advantage).

6.2 Practical Model Selection Guidance

Based on our empirical results, we outline concrete practical considerations for model selection in multi-parametric 3D medical imaging:

- **Consider 3D VisReg JEPA when**:
    1. *Annotations are scarce ($\leq 10\%$ annotated volumes)*: In data-constrained regimes (11–115 patient scans), pre-training 3D VisReg JEPA on available unannotated cohorts and fine-tuning with a UNETR hybrid decoder provides the strongest performance (34.32–57.68% mean Dice), outperforming both from-scratch ViTs ($+4.66$pp) and deeply supervised CNNs ($+7.39$pp).
    2. *Rapid frozen encoder deployment is desired*: Practitioners seeking to minimize fine-tuning compute or preserve a fixed generalist feature extractor can freeze the pre-trained ViT encoder and train only decoder heads, achieving 63.18% (pure FPN) to 69.34% (hybrid UNETR) mean Dice without backpropagating through the transformer backbone.
- **Prefer Supervised Convolutional Baselines when**:
    1. *Abundant annotations exist ($\geq 25\%$ labels)*: With extensive annotated cohorts, deeply supervised 3D nnU-Net establishes superior accuracy across all anatomical compartments (84.47% WT, 64.35% TC, 63.76% ET; mean 70.86%) and sharper boundary delineation (5.28 voxels WT, 11.66 voxels TC).
    2. *Inference latency is critical*: Standard 3D Residual UNet is over $2.7\times$ faster than UNETR Hybrid (488.82 ms vs. 1347.52 ms per 128^3 volume).

6.3 Limitations and Methodological Validity

We explicitly delineate seven methodological limitations of this study:

1. **Generative Pre-Training Comparisons**: While this study includes strictly matched from-scratch Vision Transformer controls to isolate architectural vs. pre-training effects, we do not evaluate a 3D Masked Autoencoder (3D MAE) baseline under compute-matched schedules to directly contrast latent prediction against voxel-level reconstruction. Similarly,

we evaluate VisReg as a representative single-encoder JEPA without an isolated fine-tuning comparison against SIGReg.

2. **Nature of Low-Data Background Collapse**: Standard 3D Residual UNet collapses to 4.29% mean Dice at 1% labels under the fixed 25-epoch budget (75 gradient updates at $B = 4$). This reflects an optimization failure mode under severe class imbalance and short schedules rather than an intrinsic ceiling of convolutional architectures, as deeply supervised DynUNet achieves 20.69% mean Dice under the identical budget. Furthermore, low-data sweeps evaluate fixed 25-epoch convergence without a validation checkpoint selection loop.
3. **Spatial Resolution, Canonical Coordinates, and Post-Treatment Evaluation**: Volumes are resampled from native $240 \times 240 \times 155$ ($1.0\,\text{mm}^3$) to a canonical $128 \times 128 \times 128$ grid following brain bounding box extraction. This introduces case-dependent spatial scaling (typically ≈ 1.2–1.5 mm per voxel across subjects), which is why reported HD95 surface distances are explicitly labeled in canonical grid voxels rather than physical millimeters. Regional evaluation strictly follows the official BraTS 2024 post-treatment GLI mapping (WT $= \{1, 2, 3\}$, TC $= \{1, 3\}$, ET $= \{3\}$), where label 4 denotes the surgical resection cavity (RC) and is correctly excluded from active tumor sub-compartments across all compared models.
4. **Statistical Scope and Seed Variance**: All benchmark metrics reflect a single deterministic training run (seed $= 42$). The reported standard deviations ($\pm$) capture inter-subject anatomical variance across the $N = 242$ held-out test volumes, not variance across random initializations. Multi-seed evaluations with paired bootstrap confidence intervals are needed for formal statistical hypothesis testing.
5. **Baseline Training Budgets**: The 3D nnU-Net baseline implements the MONAI DynUNet architecture trained for a matched budget of 60 epochs. Full standalone nnU-Net pipelines train for 1000 epochs with extensive automated data augmentation and ensembling, representing a substantially higher compute regime.
6. **Decoupled Projector Regularization and Uncalibrated Regularizer Weights**: In our single-encoder formulation, $\mathcal{L}_{\text{VisReg}}$ regularizes the auxiliary projector branch $g_\psi(\mathbf{Z}_{\text{ctx}}) \in \mathbb{R}^{128}$ while $\mathcal{L}_{\text{pred}}$ operates in the 384-dimensional encoder space with default weights ($\lambda_{\text{c}} = \lambda_{\text{sc}} = \lambda_{\text{sh}} = 1.0$). Because the projector can partially satisfy Gaussian moment matching using positional coordinate variance, rank is reported as a non-collapse check, and all comparative conclusions rest on downstream segmentation (Dice/HD95).
7. **Early Joint Patch Projection Bottleneck**: The 3D ViT tokenization employs a single joint linear projection across all four multi-modal sequences. While standard in UNETR designs, this topology entangles input channels and induces acute sensitivity to omitted modalities during triage. Future architectures should evaluate decoupled, per-channel patch projections paired with permutation-invariant late attention pooling or gating.

6.4 Conclusion

In this paper, we investigated whether 3D Joint-Embedding Predictive Architectures (JEPA)—designed explicitly to avoid voxel-level reconstruction—can perform dense multi-compartment volumetric medical image segmentation. Through a systematic empirical evaluation on the BraTS 2024 GLI benchmark under the three-region protocol (WT, TC, ET) comparing 3D VisReg JEPA against matched from-scratch ViT controls and supervised convolutional baselines (Residual UNet and deeply supervised DynUNet), we find that 3D JEPA successfully learns representations capable of dense multi-region segmentation, achieving $82.38 \pm 12.66\%$ WT, $59.63 \pm 37.33\%$ TC, and $59.15 \pm 36.61\%$ ET Dice (mean 67.05%) with MultiScale FPN and $84.11 \pm 12.22\%$ WT, $62.56 \pm 36.87\%$ TC, and $62.30 \pm 36.42\%$ ET Dice (mean 69.66%) when paired with a UNETR hybrid decoder. Furthermore, completely freezing the pre-trained encoder achieves 63.18% mean Dice with the pure latent FPN probe and 69.34% with the hybrid UNETR probe, showing that latent prediction preserves spatially decodable anatomical structures.

However, our empirical benchmark establishes critical architectural lessons and boundary conditions:

1. Fully supervised CNNs remain superior across all compartments on full data (84.47% WT, 64.35% TC, 63.76% ET for nnU-Net; mean 70.86%), demonstrating that localized

3D convolutional priors outperform self-attention for volumetric dense prediction when annotations are abundant.

2. Self-supervised pre-training provides decisive label-efficiency advantages in constrained annotation regimes (1–10% labels), where VisReg UNETR establishes top overall performance across the entire benchmark (34.32% at 1%, outperforming nnU-Net by +13.63pp; and 54.45% at 5%, outperforming scratch ViT by +4.66pp and nnU-Net by +7.39pp).
3. VisReg JEPA establishes superior robustness on emergency single-sequence triage, reaching top performance on FLAIR-only acquisitions (44.61% vs. 39.79% for nnU-Net) and shielding transformers against severe degradation on T1c-only (49.22% vs. 36.55% for scratch ViT).

We conclude that 3D JEPA provides a powerful representation foundation for volumetric medical imaging under annotation scarcity, delivering high spatial decodability and robust multi-compartment segmentation without pixel-level reconstruction.

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